

ARNAV SINGH

Software Engineer

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SUMMARY

AI Systems and full-stack engineer building production-oriented platforms across trading systems, geospatial safety engines, biosignal/EEG analysis, and developer tooling. Experienced with FastAPI, Next.js, TypeScript, and Python across multi-service, event-driven architectures with RAG and LLM integration.

EDUCATION

Chandigarh University Uttar Pradesh

Uttar Pradesh, India

Bachelor of Technology, Computer Science and Engineering | CGPA: 7.15

July 2025 – July 2029

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Backend & APIs: FastAPI, Express.js, Spring Boot, RESTful APIs, WebSockets

Frontend: React 18, Next.js 14/15, Vite

Databases & Data: PostgreSQL, PostGIS, pgvector, TimescaleDB, Redis, SQLAlchemy, Pydantic

AI / ML: RAG Pipelines, LLM Orchestration, Keras/CNN Models, Agentic Workflows

Cloud & DevOps: Docker, Vercel, Firebase Auth, Git/GitHub, CI/CD (GitHub Actions)

Other: JWT/HMAC-SHA256 Auth, AES-256-GCM Encryption, Microservices Architecture

PROJECTS

SIRUS — Multi-Tenant Automated Trading Platform

Python, FastAPI, Next.js 14, TimescaleDB, Redis, Firebase Auth | github.com/arnnnnaavvvvv/SIRUS

- Designed an event-driven microservices architecture separating market-data ingestion, strategy evaluation, risk validation, and order execution via a Redis pub/sub backbone.
- Built a broker adapter layer unifying Alpaca and KuCoin APIs with a pluggable strategy engine, historical backtester, atomic tenant locking, and AES-256-GCM credential encryption.

IGNITE — AI-Powered Tourist Safety & Smart Route Planner

Next.js/Vite, React 18, FastAPI, PostGIS, Leaflet.js, Redis | github.com/arnnnnaavvvvv/IGNITE

- Built a pan-India, destination-agnostic safety engine resolving free-text locations to coordinates and elevation, with a deterministic multi-factor risk scoring system across 6 terrain profiles.
- Implemented dynamic re-routing against live hazard polygons and weather alerts, an SOS dispatch system for nearby emergency services, and polished UI/UX (Framer Motion); shipped live and publicly deployed.

NeuroSense — EEG-Based Early-Warning System

Next.js 14, FastAPI, PostgreSQL (pgvector), Redis

- Built an EEG signal classification system using a pretrained CNN (CHB-MIT dataset) originally developed for epileptic seizure-risk staging, now expanded toward stress and anxiety early-warning detection.
- Layered a RAG-based retrieval pipeline on top of CNN output to surface grounded, context-aware precaution guidance, sharing one CNN backbone across both detection variants.

CLUDE — Git-Commit Root-Cause Analysis & Onboarding Assistant

Next.js 14, FastAPI, PostgreSQL (pgvector), Redis

- Built a developer tool combining git-commit root-cause analysis with an AI assistant that helps new engineers onboard onto an unfamiliar codebase.
- Designed semantic causal reasoning over git history as the core differentiator — surfacing why a change was made, not just which line was last touched — with full documentation including Mermaid architecture diagrams.

ACHIEVEMENTS

- Top 15 Finalist, IEEE Idea2Impact 2026 — National Innovation Challenge (IEEE Computational Intelligence Society), as part of Team Digital Destroyer, Chandigarh University.